MARK **HINTZ**

MECHANICAL DESIGNER • SOLIDWORKS • MACHINE SHOP • SHEET METAL • DTO MANUFACTURING • PDM • AI DEV TOOLS

Jacksonville, FL
markworks.dev@gmail.com
LinkedIn: linkedin.com/in/markhintz
(904) 862-1945

EXPERTISE

• CAD & Design
SolidWorks (Advanced), PDM Vault, AutoCAD, Inventor, assemblies, weldments, sheet metal, GD&T

• Documentation
2D fabrication drawings, DXF creation, layer control, BOMs, exploded views, revision control, standards

• Manufacturing & DFM
Laser cutting, press brake, CNC prep, jigs/fixtures, tolerance analysis, design for manufacturability & assembly

• Design-to-Order (DTO)
Configurable platforms, reuse of common components, options management, drawing/BOM variants

• Collaboration
Close coordination with engineers, manufacturing, and shop personnel; support for build issues and installation

• Process & Automation
Basic Python/JavaScript, internal tools to streamline BOM work, drawing checks, and scheduling.

Education

A.S. — Computer-Aided Drafting & Design
ITT Technical Institute
Jacksonville, FL (2002)

PROFILE

Mechanical Designer with 15+ years of experience in 3D modeling, 2D fabrication drawings, and BOM management for custom, design-to-order industrial equipment. Strong background in SolidWorks, sheet metal, structural frames, and production documentation that shop and fabrication teams can trust. Experienced in building multi-variant product families from common components (DTO) and standardizing designs for manufacturability. Known for tightening up workflows, improving manufacturability, and using light automation to reduce errors and lead times while keeping designs practical and buildable.

PROFESSIONAL EXPERIENCE

MECHANICAL DESIGNER / AI-DRIVEN TOOLS DEVELOPER

MYERS-SETH PUMPS (MSP) — Jacksonville, FL

- Design complete pump packages, engine mounts, sheet-metal components, lifting structures, vacuum systems, and chassis in SolidWorks within a design-to-order environment.
- Configure base platforms into customer-specific variants (models, engines, control panels, trailer options), then deliver accurate 3D models, 2D fabrication drawings, and BOMs for each order.
- Produce fabrication drawings, DXFs, and flat patterns for laser cutting and brake forming, with clear datums, weld symbols, and tolerances.
- Create and maintain accurate Bills of Materials (BOMs) for complex assemblies; standardize naming, part numbering, and revision practices to reduce errors across variants.
- Work directly with fabrication, weld, and assembly teams to resolve build issues, improve fit-up, and refine designs for manufacturability and serviceability.
- Built “PumpTracker,” an internal production scheduling and capacity planning tool to visualize workload across Fabrication, Powder Coat, Assembly, and Testing.
- Developed internal tools that assist with BOM generation, drawing QA, and workflow diagnostics—reducing manual rework and bottlenecks.

MECHANICAL DESIGNER / AI-DRIVEN TOOLS DEVELOPER

SPECIAL TOOL SOLUTIONS / BCP — Jacksonville, FL

- Designed pneumatic planetary reduction gearboxes, handle assemblies, and related hardware for industrial torque tools across multiple product lines each with several torque/output/RPM variants.
- Started from OEM-derived drawings and early prototypes, then iterated and improved designs for manufacturability, durability, and ease of assembly; eventually took on design control for new DTO variants in the product families.
- Created detailed 3D models, 2D assembly drawings, exploded views, and tolerance-controlled prints to support repeatable production.
- Reduced prototype runs from 3-4 units to near zero by tightening documentation, clarifying critical dimensions, and aligning designs with shop capabilities.
- Created the tools and databases used to manage projects from quote through production: scheduling, materials, vendor coordination, and production paperwork.
- Used 3D scanning and tools like Geomagic to reverse-engineer real-world geometry into accurate, parametric CAD models.
- Built custom Excel-based databases to track job status, parts, and throughput across machining and assembly.